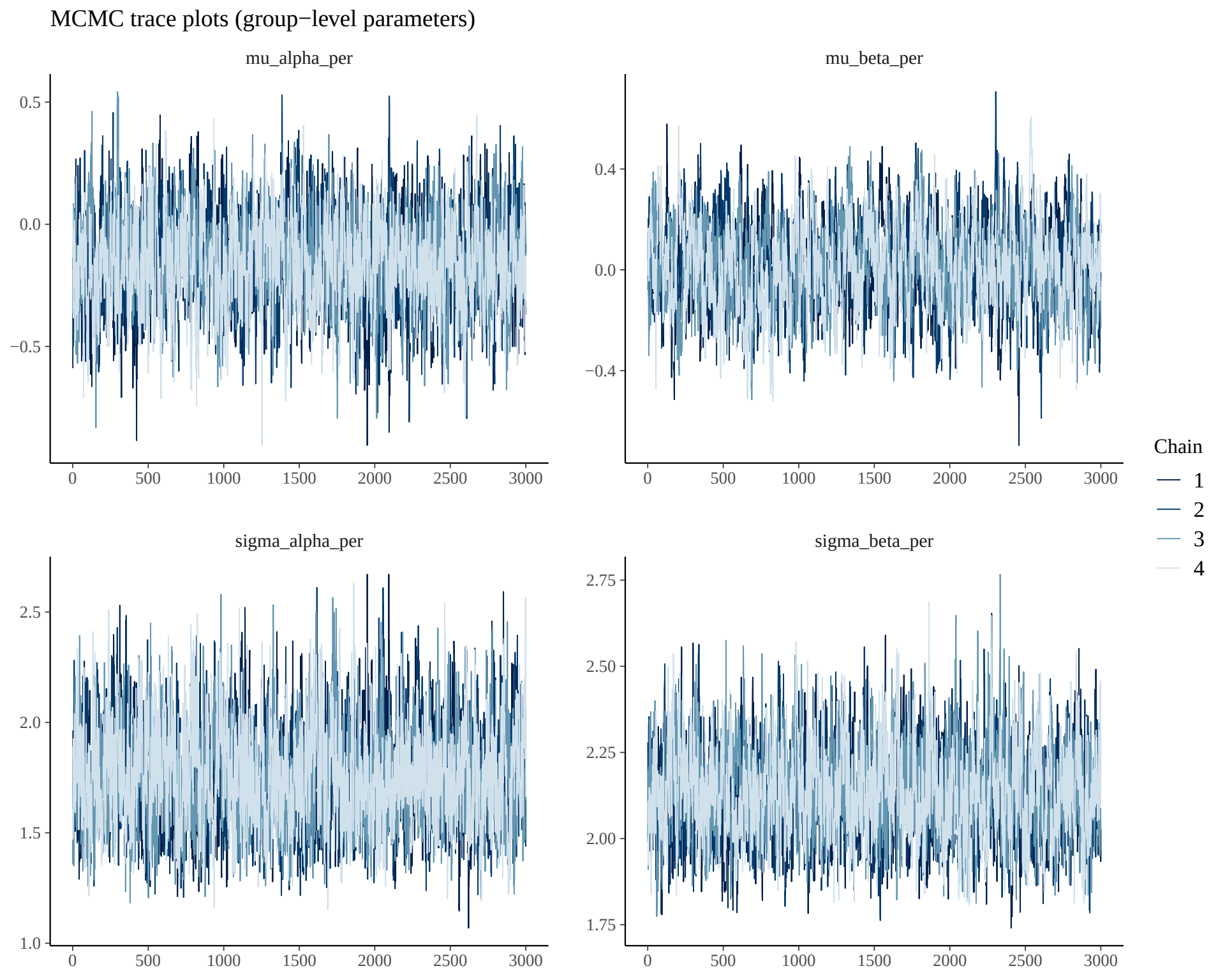
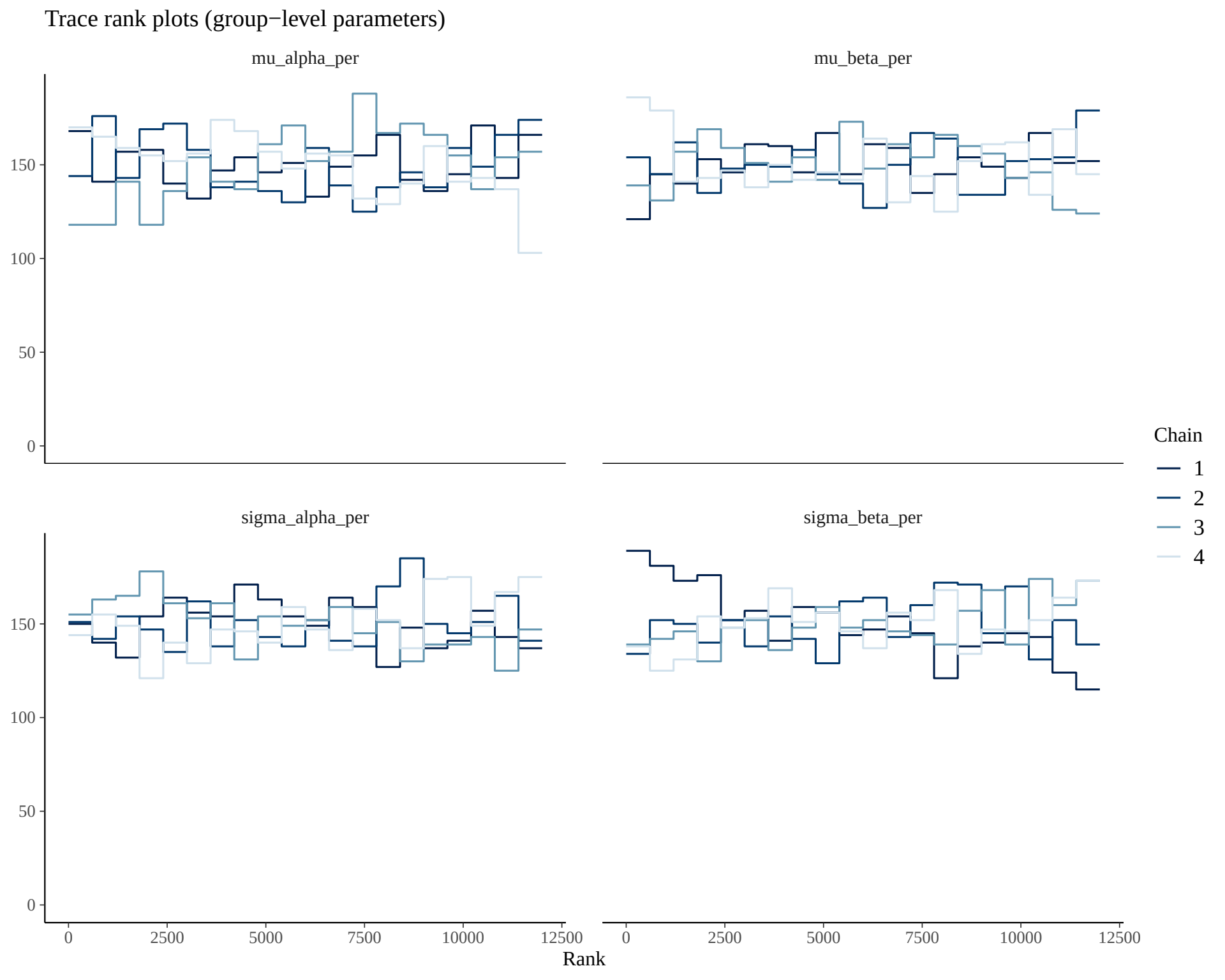
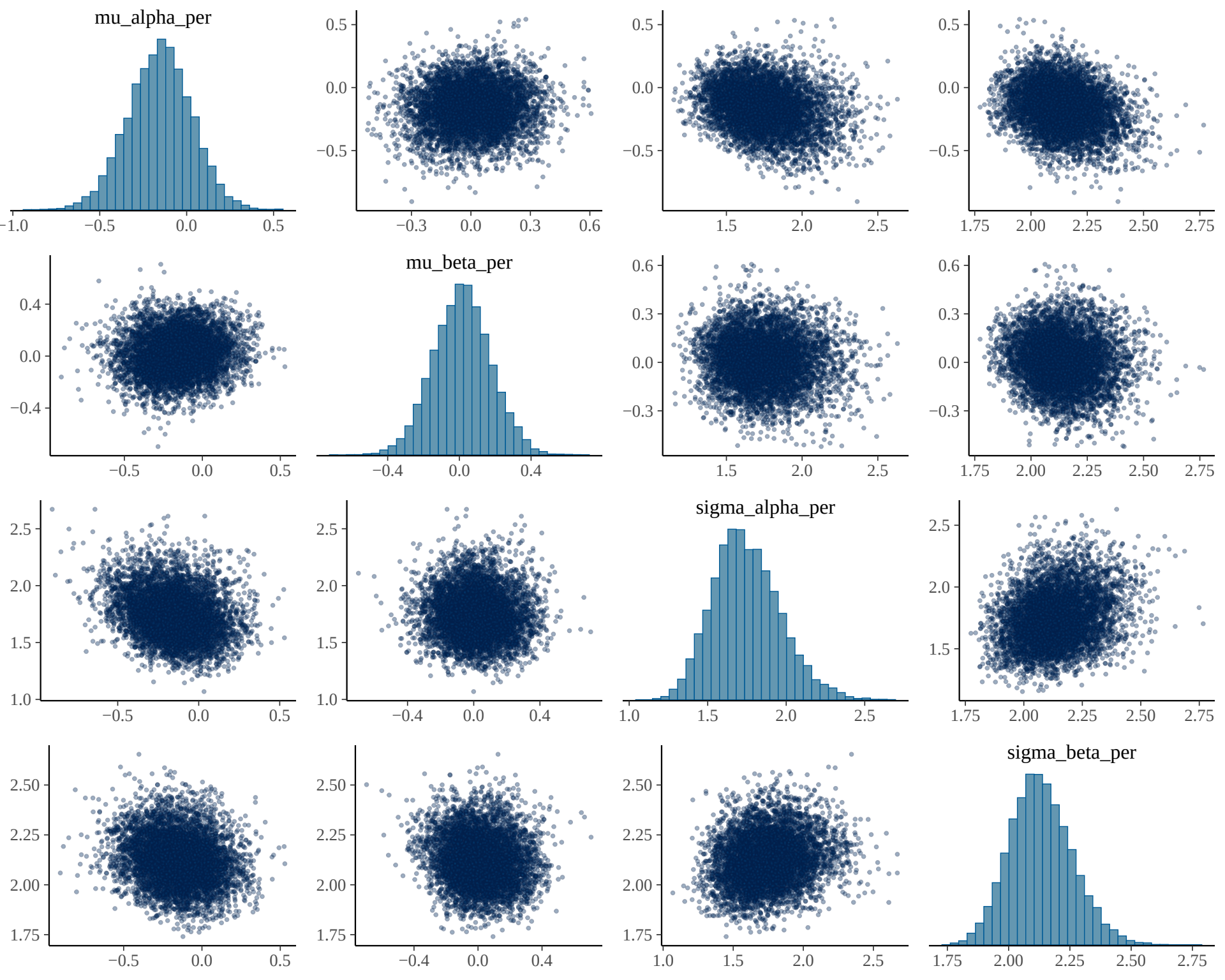
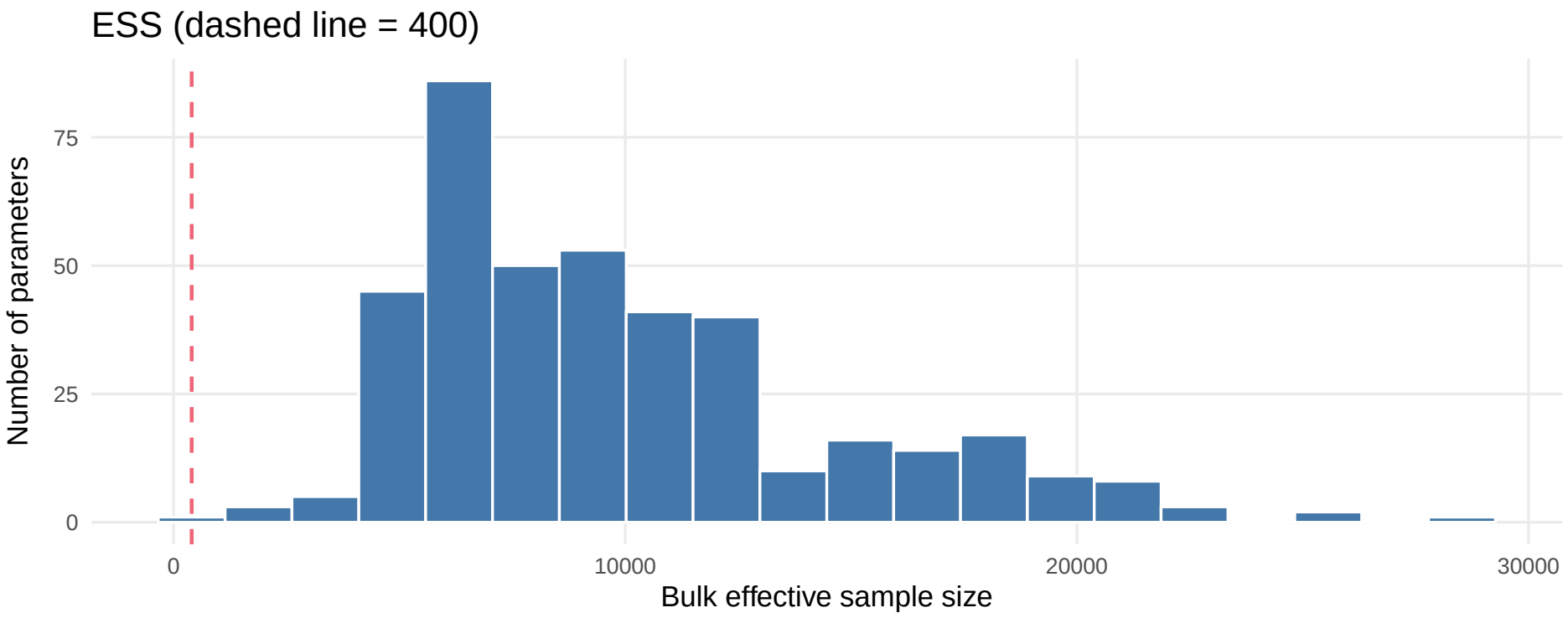
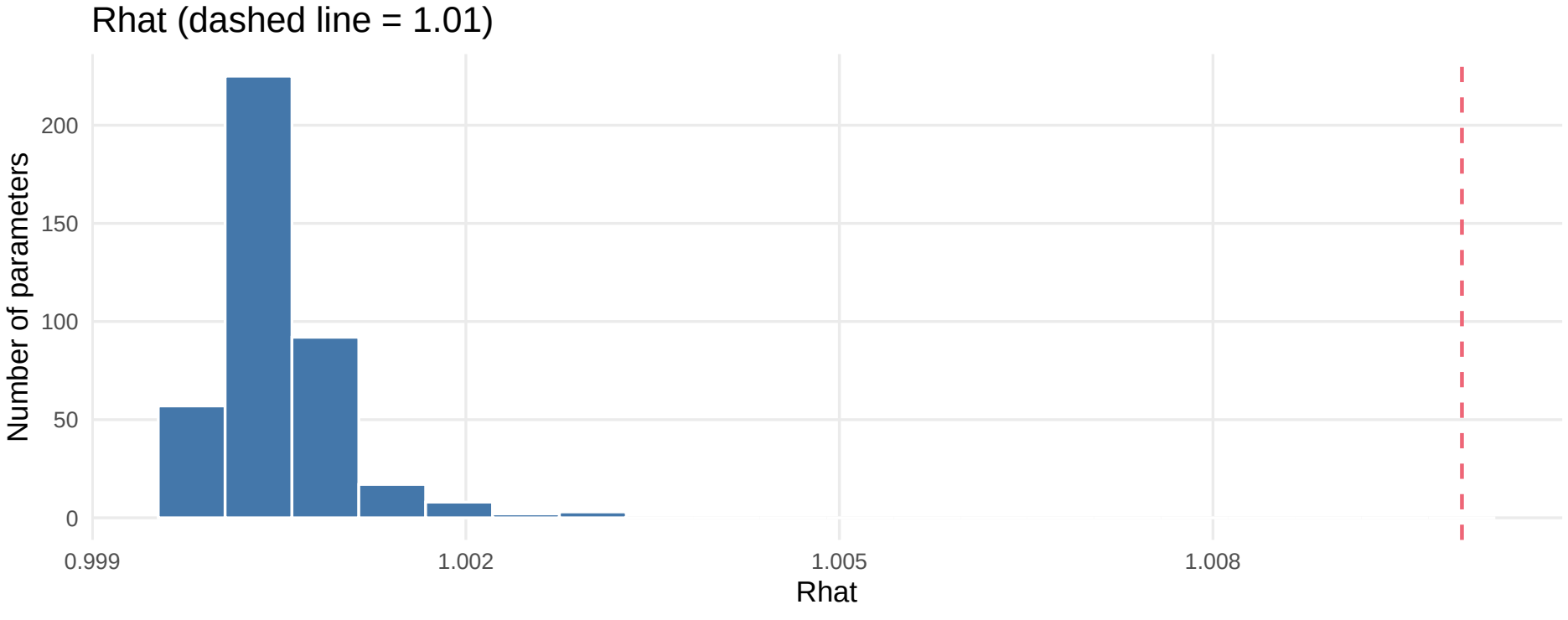

MCMC trace plots (group−level parameters)
mu_alpha_per
mu_beta_per
sigma_alpha_per
sigma_beta_per
0.5
0.0
−0.5
0.4
0.0
−0.4
2.5
2.0
1.5
1.0
2.75
2.50
2.25
2.00
1.75
0
500
1000
1500
2000
2500
3000
Chain
1
2
3
4


Trace rank plots (group−level parameters)
mu_alpha_per
mu_beta_per
sigma_alpha_per
sigma_beta_per
0
50
100
150
0
2500
5000
7500
10000
12500
Rank
Chain
1
2
3
4


mu_alpha_per
mu_beta_per
sigma_alpha_per
sigma_beta_per


ESS (dashed line = 400)
Number of parameters
75
50
25
0
0
10000
20000
30000
Bulk effective sample size


Rhat (dashed line = 1.01)
Number of parameters
200
150
100
50
0
0.999
1.002
1.005
1.008
Rhat